

GPT-OSS-120B VIS Benchmark

without aiDAPTIV vs with aiDAPTIV (Cold / Warm)

True GPT-OSS-120B NP1-32 serving data. Same visual logic as the 70B report, with Cold and repeated-prompt Warm separated.

1) Benchmark scope

Model	gpt-oss-120b
NP levels	1 / 2 / 4 / 8 / 16 / 32
Input tokens	1K / 2K / 4K / 8K / 16K
Output tokens	1K / 2K

2) Comparison modes

without aiDAPTIV	3-run baseline average (Run1/2/3)
with aiDAPTIV (Cold)	Run1
with aiDAPTIV (Warm)	Average of Run2 and Run3

3) Metrics in this report

TTFT (ms)	Time from request start to first token
TPOT (ms/token)	Median TPOT based decode timing
Aggregate throughput	Raw vLLM Output token throughput (tok/s)
Normalized throughput	Aggregate throughput / NP (tok/s/user)

4) Reading guide

TTFT / TPOT	Lower is better
Throughput	Higher is better
Warm state	Repeated same-prompt runs on the same live server
TPOT robustness	Median TPOT; invalid short-output warm runs filtered for TPOT only

Method note: Warm = repeated-prompt Run2/Run3 behavior. The runner remained live; this is not the formal random_run_phase=2 80% reuse mode.

TTFT

Time to first token (ms). Lower is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

Without aiDAPTIV

Cold (Run1)

Warm (Run2/Run3 avg)

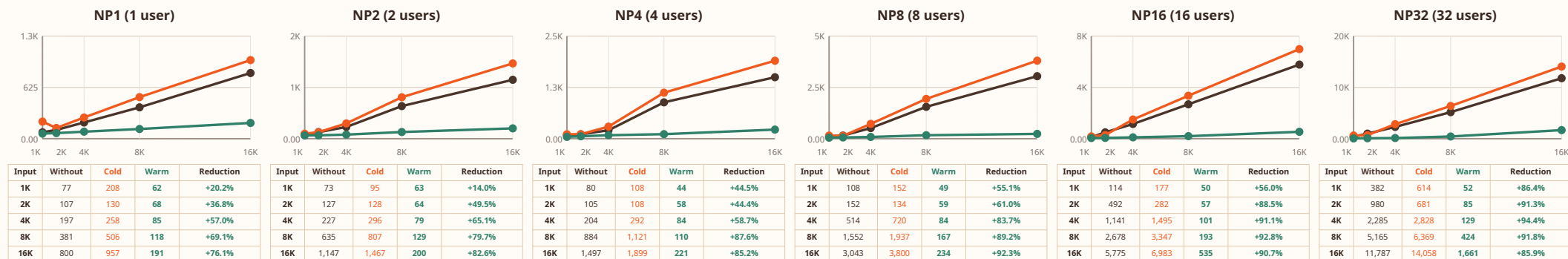
FORMULA

TTFT (ms) = mean(first-token time - request-start time)

Reduction (%) = (Without - Warm) / Without x 100

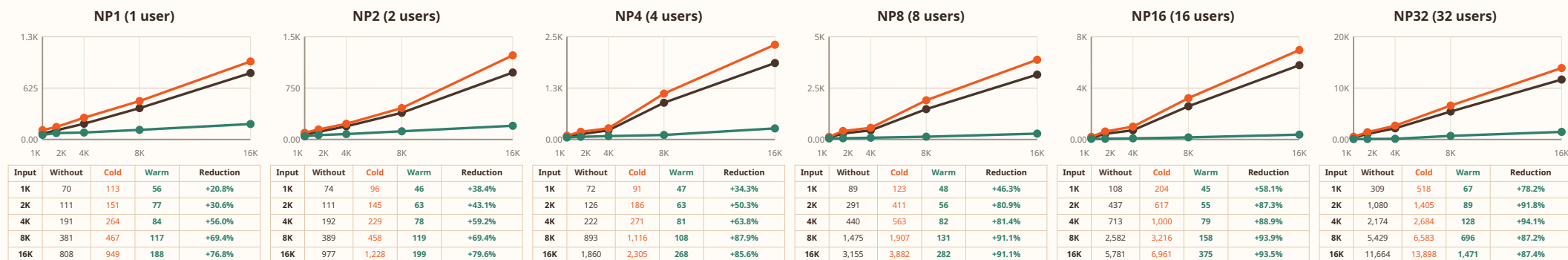
Output 1K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



Output 2K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



KEY TAKEAWAY TTFT improves from NP1 / Input 1K and meets the 50% POC goal from Input 4K, peaking at 94.4%.

TPOT

Median time per output token (ms/token). Lower is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

Without aiDAPTIV

Cold (Run1)

Warm (Run2/Run3 avg)

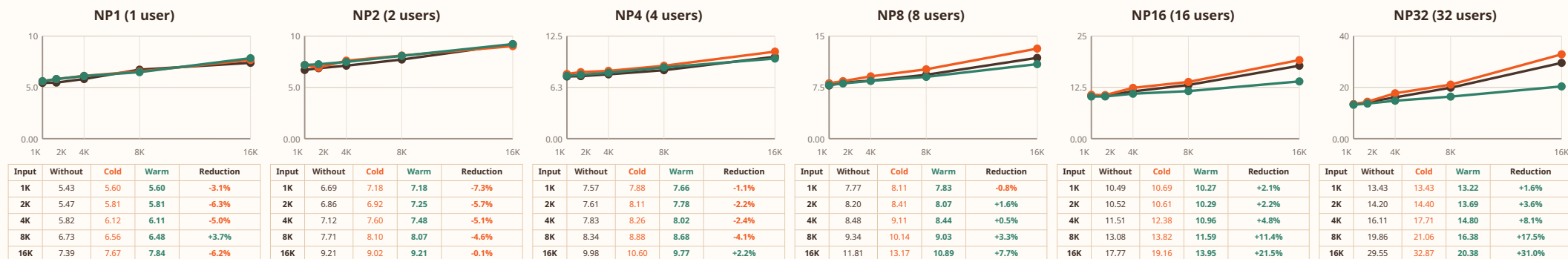
FORMULA

$$\text{TPOT} = (\text{E2E latency} - \text{TTFT}) / (\text{output tokens} - 1)$$

$$\text{Reduction (\%)} = (\text{Without} - \text{Warm}) / \text{Without} \times 100$$

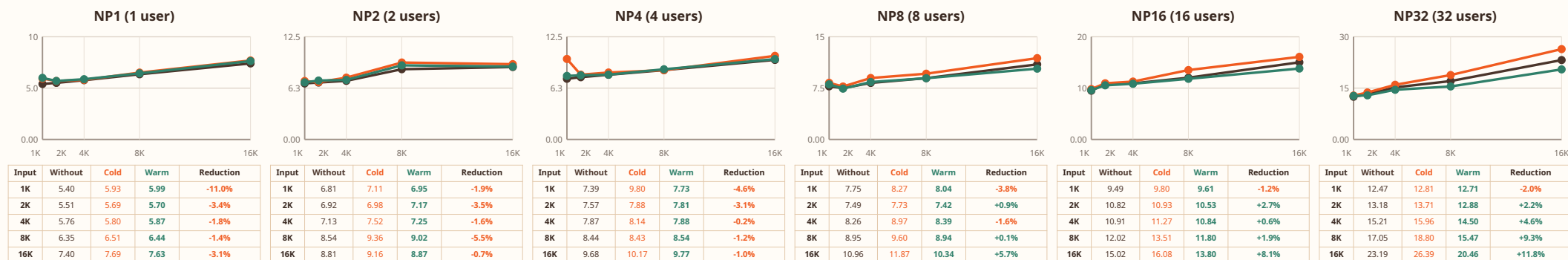
Output 1K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



Output 2K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



KEY TAKEAWAY TPOT improves from NP16 / Input 2K but misses the 50% POC goal, peaking at 31.6%.

Throughput (Aggregate)

Raw vLLM output token throughput (tok/s). Higher is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

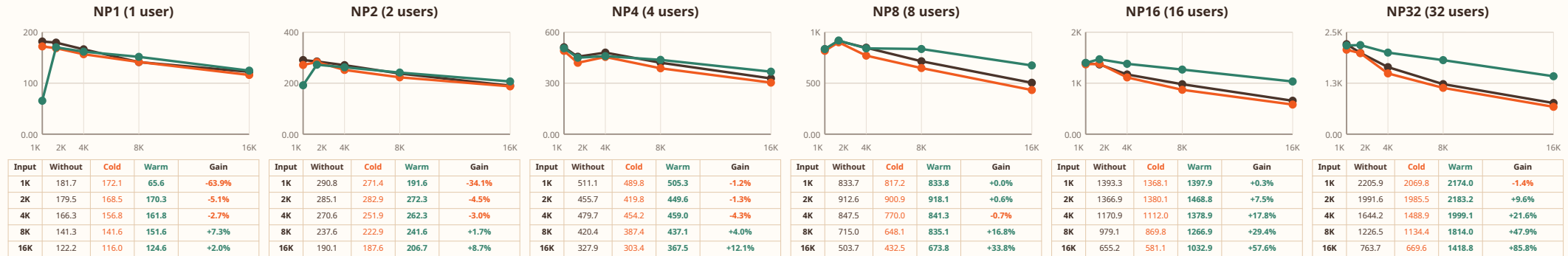
FORMULA

Aggregate tok/s = total generated output tokens / benchmark duration

Gain (%) = (Warm / Without - 1) x 100

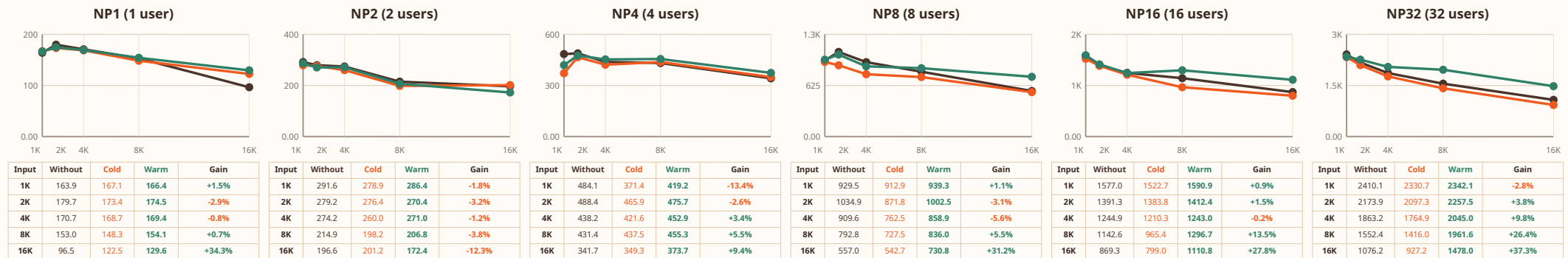
Output 1K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



Output 2K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



KEY TAKEAWAY Aggregate throughput improves from NP16 / Input 2K and meets the 50% POC goal at Input 16K / Output 1K, peaking at 85.8%.

Throughput (Normalized)

Aggregate throughput divided by NP (tok/s/user). Higher is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

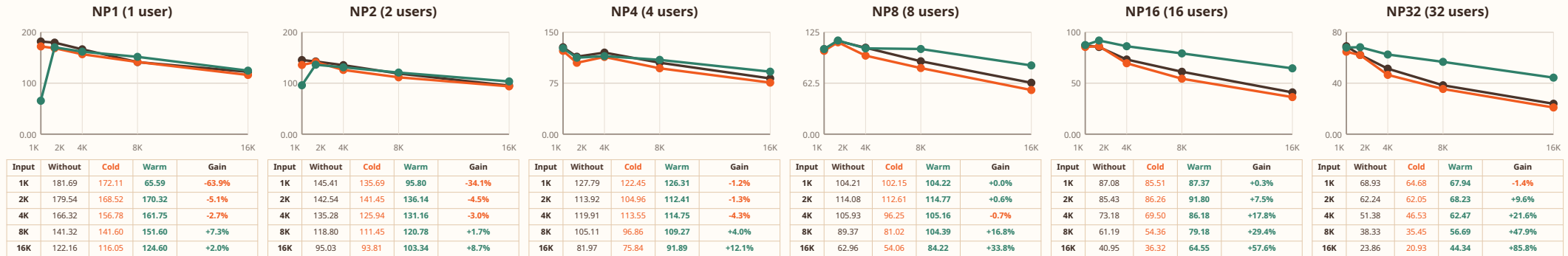
FORMULA

Normalized tok/s/user = Aggregate tok/s / NP

Gain (%) = (Warm / Without - 1) x 100

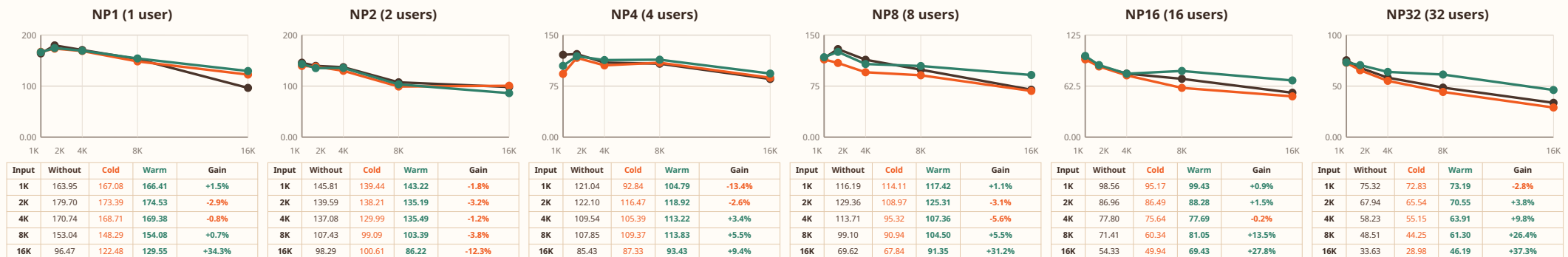
Output 1K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



Output 2K

Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.



KEY TAKEAWAY Per-user throughput improves from NP16 / Input 2K and meets the 50% POC goal at Input 16K / Output 1K, peaking at 85.8%.

SUMMARY

aiDAPTIV impact on latency, decode efficiency, and throughput

GPT-OSS-120B VIS benchmark, NP1-32; summary highlights target attainment and 16K long-context scaling.

1

TTFT (Time to First Token)

>50% POC target achieved from 4K input.

Overall maximum

94.4% lower

16K long-context reduction

NP16

90.7%-93.5%

NP32

85.9%-87.4%

2

TPOT (Time per Output Token)

Improves at high NP / long context, but 50% target is not reached.

NP32 / 16K / Output 1K

29.55 -> 20.38

-31.0%

Output 2K

23.19 -> 20.46

-11.8%

3

Aggregate Throughput (tok/s)

16K / Output 1K exceeds the 50% target at NP16 and NP32.

NP16 / 16K / Output 1K

655.2 -> 1032.9

+57.6%

NP32 / 16K / Output 1K

763.7 -> 1418.8

+85.8%

4

Normalized Throughput (tok/s/user)

Per-user capacity shows the same 16K scaling pattern.

NP16 / 16K / Output 1K

40.95 -> 64.55

+57.6%

NP32 / 16K / Output 1K

23.86 -> 44.34

+85.8%

KEY TAKEAWAYS

1

TTFT >50% target achieved

94.4% lower

maximum reduction

2

TPOT 50% target not reached

31.0% lower

best observed reduction

3

Throughput >50% achieved

+85.8%

NP32 / 16K / Out1K